

Mar 26 - Math Class

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Topic

Symmetry in Geometry: Discovering Line Symmetry through Movement and Art

Student :

Grade 5 (U.S. K-12)

Class size: 15 students

Duration: 40 minutes

Learning Outcomes: SWBAT

1. **Identify** what a line of symmetry is.
2. **Recognize** whether a shape or picture is symmetrical.
3. **Create** a simple symmetrical figure using art, movement, and paper cutting.
4. **Apply** the idea of symmetry in art.

Materials for the class:

- Slides of symmetrical objects in daily life
- Tape to divide the classroom floor into two parts
- Music by Bach related to symmetry / mirrored structure
- Colorful paper
- Paint or watercolor
- Brushes
- Scissors

Time	Activity	purpose
5 min	Warm-up / Introduction Teacher Actions ● Show students several familiar pictures from daily life, such as butterflies, leaves, hearts, masks, and buildings. ● Ask simple questions. ● Introduce the term line symmetry. ● Explain that a symmetrical figure has two sides that match across a line. Student Actions ● Observe pictures. ● Share ideas about which pictures look “the same on both sides.” ● Learn the meaning of symmetry and line of symmetry.	To activate prior knowledge and introduce the main geometric concept in a visual and simple way.
10 min	Activity 1:Symmetry through Music and Movement Teacher Actions ● Put tape on the classroom floor to divide the room into two parts. ● Tell students to imagine the tape is the line of symmetry. ● Divide students into two groups. One group stands on the left side. The other group stands on the right side. ● Play Bach's music. ● Students on the left move freely with simple guided movement prompts. ● Teacher says “Freeze!” ● Students on the right observe and copy the frozen positions as mirrored partners. Student Actions ● Listen to music and move.	To let students physically experience the idea of symmetry, mirror image, and the line of symmetry.

	● Freeze when they hear the instruction. ● Mirror their classmates' body positions. ● Reflect on whether the movement is symmetrical.	
8 min	Activity 2: Fold Painting / Symmetry Art (8 minutes) Teacher Actions: ● Distribute folded paper. ● Instruct students to paint on one side. ● Ask them to fold, press, and open the paper. ● Guide students to observe the symmetrical pattern. Student Actions ● Paint on one side of the folded paper. ● Fold and press the paper. ● Open it and observe the symmetrical image.	Utilize the axis of symmetry of origami.
8 min	Activity 3: Symmetrical Drawing with Hands and Partner Work (8 minutes) Teacher Actions ● First, ask students to try drawing a symmetrical figure using both hands at the same time. ● Then put students in pairs. ● Each pair gets one paper with a middle line.	To practice spatial awareness, precision, and cooperation while creating symmetry.

	● One student draws half of a picture on one side, and the other student tries to draw the matching half on the other side. ● Encourage communication and teamwork. Student Actions ● Try using both hands to create a symmetrical drawing. ● Work with a partner to complete a matching symmetrical image. ● Compare the two sides and adjust if needed.	
6 min	Activity 4: Paper Cutting Final Product (6 minutes) Teacher Actions ● Demonstrate how to fold paper and cut a simple symmetrical shape. ● Ask students to create their own symmetrical paper-cut design as a final product. ● Invite a few students to share and point out the line of symmetry. Student Actions ● Fold paper. ● Cut a symmetrical figure. ● Show their work and explain where the line of symmetry is.	To give students a final hands-on product that demonstrates their understanding of symmetry.

Vocabulary:

- symmetry
- line of symmetry
- mirror

- same
- match
- left
- right
- fold
- shape
- cut

The axis of symmetry 对称轴 is a special line that divides a shape into two equal halves, like a mirror. It's like folding a shape in half and the two sides match up perfectly. Different shapes have different numbers of lines of symmetry. A square has four lines of symmetry, a rectangle has two, and a circle has an infinite number of lines of symmetry.

In summary, the axis of symmetry is a line that divides a shape into two equal halves.

axis (noun) 轴

an imaginary line around which an object rotates or is symmetrically arranged.

An axis is a line in the middle of something.

symmetry (noun) 对称

Symmetry means both sides look the same.

One side is like the mirror image of the other side.